

Learning Outcome: Write C Programming Codes

Total Marks: _____ / 70

Trainee Name: _____

Date: _____

PART I — Choose the Correct Answer

- Which of the following is NOT a token in C?
A. Keyword B. Variable C. Operator D. Function prototype
- Which of these is a valid C data type?
A. integer B. char C. float32 D. string
- Which of the following is a correct constant declaration?
A. `int const x = 10;` B. `constant int x = 10;` C. `x = 10 const int;` D. `int x const = 10;`
- Which operator is used for logical AND in C?
A. & B. && C. | D. ||
- Which of the following is NOT a preprocessor directive?
A. `#define` B. `#include` C. `#pragma` D. `#function`
- In C, the entry point of a program is:
A. `void main()` B. `main()` C. `int main()` D. `program()`
- Which statement correctly declares a function returning an integer?
A. `int add(int a,int b);` B. `function int add(...)` C. `int function add` D. `add(...) int`
- Which of these is a single-line comment in C?
A. `/* */` B. `// comment` C. `<!-- -->` D. `# comment`
- What will the following C code print if **x = 10**?

```
if (x > 5){  
    printf("Hello");  
    }else{  
    printf("world");  
    }
```


A. Hello B. world C. Error D. Nothing
- What is the return type of the function shown below?

```
float multiply(float a, float b){  
    return a * b;  
}
```


A. int B. void C. float D. double

PART II — True or False

- Function arguments are values passed into a function when called.
- `#include <stdio.h>` is NOT a preprocessor directive.
- % is a modulus arithmetic operator.
- `string s = "Hello";` is valid in standard C.
- A function declaration specifies name, return type, parameters.
- `int sum(int,int);` is a valid declaration.
- Comments affect execution.
- switch can replace many else-if checks.
- void function returns a value.
- The following code will print "Yes". Study the program below:

```
int x = 5;  
  
if(x < 10){  
    printf("Yes");  
}else if(x < 20){  
    printf("Maybe");  
}else  
    printf("No");
```

PART III — Structured Questions

- Sequence these C program components in typical order:
 - ___ . Functions
 - ___ . Preprocessor directives
 - ___ . Statements & expressions
 - ___ . Comments
- Order correctly these steps of writing a C function:
 - ___ . Call the function
 - ___ . Define the function
 - ___ . Declare the function
- Match the following C keywords/functions with their correct description:

No	Item	Match(Fill in the Correct item)	Description
1	float		Data type for integers
2	void		Decimal numbers
3	switch		Single characters
4	int		No return value
5	else if		Decision statement
6	if		New condition
7	char		Multiple cases
8	#define		Define constants
9	%		Output function
10	printf()		Remainder operator

PART IV — Program Creation

- Write a complete C program that asks the user to enter a number and checks whether the number is **positive, negative, or zero**. The program must display the correct message based on the value entered.
- Write a complete C program that asks the user to enter a number from **1 to 7** and prints the corresponding **day of the week** using a **switch statement**. Display an error message if the number is outside the valid range.
- Write a C program that asks the user to enter a number and determines whether the number is **even or odd** using a **user-defined function**. The function should return a value indicating whether the number is even or odd, and the main program should print the result.
- Write a C program that reads a **student's name** and **marks**, then determines and prints the student's **grade using a user-defined function**. The function should return the grade based on the grading rules below. Output must look like this:

```
Student: Alice
Marks: 78
Grade: B
```

Grading Rules	
A	85 – 100
B	70 – 84
C	55 – 69
D	40 – 54
Fail	< 40